FACULTY OF COMPUTING & DATA SCIENCE

Section 03 | Review of sections 01 + 02

# Section 1,2 REVIEW

Decimal to binary · Computers · Number systems

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SECTION 03 · TODAY

# THE PLAN

01

## Decimal to binary

Whole numbers and fractions.

02

## Intro to computers

A short recap of the key ideas.

03

## Number systems

Binary, octal, decimal and hex.

04

## Practice round

Two questions, then the answers.

PART 01

# DECIMAL TO BINARY

Split at the point. Each side has its own rule.

PART 01 · THE METHOD

# TWO PARTS, TWO RULES

INTEGER PART · BEFORE THE POINT

**152**

↑ **Divide by 2, read the remainders up.**

FRACTION PART · AFTER THE POINT

**.125**

↓ **Multiply by 2, read the integer digits down.**

WORKED EXAMPLE · 152.125 · STEP 1

# INTEGER: 152

| STEP | QUOTIENT | BIT |
|---|---|---|
| 152 ÷ 2 | 76 | 0 |
| 76 ÷ 2 | 38 | 0 |
| 38 ÷ 2 | 19 | 0 |
| 19 ÷ 2 | 9 | 1 |
| 9 ÷ 2 | 4 | 1 |
| 4 ÷ 2 | 2 | 0 |
| 2 ÷ 2 | 1 | 0 |
| 1 ÷ 2 | 0 | 1 |



READ THE REMAINDERS UP

**10011000**

**The last remainder is the first bit.**

WORKED EXAMPLE · 152.125 · STEP 2

# FRACTION: .125

| STEP | RESULT | BIT |
|---|---|---|
| 0.125 × 2 | 0.25 | **0** |
| 0.25 × 2 | 0.5 | **0** |
| 0.5 × 2 | 1.0 | **1** |

**Stop: the fraction is now 0.**

READ THE DIGITS DOWN

## .001

**The first digit goes right after the point.**

ANSWER · 152.125 IN BINARY

# $(10011000.001)_2$

| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | | 0.5 | 0.25 | 0.125 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | . | 0 | 0 | 1 |

**INTEGER PART**

**128 + 16 + 8 = 152**

**FRACTION PART**

**1/8 = 0.125**

EXAMPLE 1

# EXAMPLE 1: 25.375

INTEGER 25 · READ UP

| STEP | QUOTIENT | BIT |
|---|---|---|
| 25 ÷ 2 | 12 | 1 |
| 12 ÷ 2 | 6 | 0 |
| 6 ÷ 2 | 3 | 0 |
| 3 ÷ 2 | 1 | 1 |
| 1 ÷ 2 | 0 | 1 |

FRACTION .375 · READ DOWN

| STEP | RESULT | BIT |
|---|---|---|
| 0.375 × 2 | 0.75 | 0 |
| 0.75 × 2 | 1.5 | 1 |
| 0.5 × 2 | 1.0 | 1 |

ANSWER

$(11001.011)_2$

EXAMPLE 2

# EXAMPLE 2: 43.6875

INTEGER 43 · READ UP

| STEP | QUOTIENT | BIT |
|---|---|---|
| 43 ÷ 2 | 21 | 1 |
| 21 ÷ 2 | 10 | 1 |
| 10 ÷ 2 | 5 | 0 |
| 5 ÷ 2 | 2 | 1 |
| 2 ÷ 2 | 1 | 0 |
| 1 ÷ 2 | 0 | 1 |

FRACTION .6875 · READ DOWN

| STEP | RESULT | BIT |
|---|---|---|
| 0.6875 × 2 | 1.375 | 1 |
| 0.375 × 2 | 0.75 | 0 |
| 0.75 × 2 | 1.5 | 1 |
| 0.5 × 2 | 1.0 | 1 |

ANSWER

$(101011.1011)_2$

EXAMPLE 3 · AN IMPORTANT CASE

# EXAMPLE 3: 10.3

INTEGER 10 · READ UP

| STEP | QUOTIENT | BIT |
|---|---|---|
| 10 ÷ 2 | 5 | 0 |
| 5 ÷ 2 | 2 | 1 |
| 2 ÷ 2 | 1 | 0 |
| 1 ÷ 2 | 0 | 1 |

FRACTION .3 · READ DOWN

| STEP | RESULT | BIT |
|---|---|---|
| 0.3 × 2 | 0.6 | 0 |
| 0.6 × 2 | 1.2 | 1 |
| 0.2 × 2 | 0.4 | 0 |
| 0.4 × 2 | 0.8 | 0 |
| 0.8 × 2 | 1.6 | 1 |

ABOUT · STOP AT 5 DIGITS

$(1010.01001)_2$

0.6 is back. It never reaches 0.

Your turn

5 minutes

CONVERT TO BINARY

# $(231.0625)_{10}$

Write both tables, then check by converting back.

STUCK ON 231.0625?

# ONE HINT AT A TIME

Click for the next hint

Hint 1

Split it: 231 and 0.0625.

Hint 2

Try it yourself first

Hint 3

Try it yourself first

SOLUTION

# SOLUTION: 231.0625

INTEGER 231 · READ UP

| STEP | QUOTIENT | BIT |
|---|---|---|
| 231 ÷ 2 | 115 | 1 |
| 115 ÷ 2 | 57 | 1 |
| 57 ÷ 2 | 28 | 1 |
| 28 ÷ 2 | 14 | 0 |
| 14 ÷ 2 | 7 | 0 |
| 7 ÷ 2 | 3 | 1 |
| 3 ÷ 2 | 1 | 1 |
| 1 ÷ 2 | 0 | 1 |

FRACTION .0625 · READ DOWN

| STEP | RESULT | BIT |
|---|---|---|
| 0.0625 × 2 | 0.125 | 0 |
| 0.125 × 2 | 0.25 | 0 |
| 0.25 × 2 | 0.5 | 0 |
| 0.5 × 2 | 1.0 | 1 |

ANSWER

$$(11100111.0001)_2$$

Check: 128 + 64 + 32 + 4 + 2 + 1 = 231, and 1/16 = 0.0625.

PART 01 · WATCH OUT

# COMMON MISTAKES

## Remainders read down

✗ `152 → 00011001`

✓ `152 → 10011000`

Read the remainders from the bottom up.

## Fraction read up

✗ `.125 → .100`

✓ `.125 → .001`

Read the fraction digits from the top down.

## Keeping the 1

✗ `1.5 × 2 = 3.0`

✓ `0.5 × 2 = 1.0`

Take the 1 out. Continue with the fraction only.

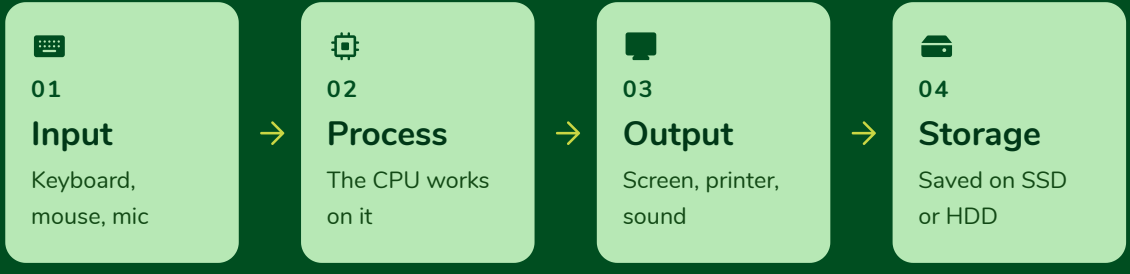
PART 02 · SECTION 01 RECAP

# INTRO TO COMPUTERS

Only the key ideas, fast.

PART 02 · HOW IT WORKS

# THE IPOS CYCLE




**Example** **Type 2 + 3, the CPU adds, 5 shows, history saves it.**

PART 02 · BASICS

# HARDWARE VS SOFTWARE



## Hardware

The parts you can touch.

CPU, RAM, keyboard, monitor, SSD.



## Software

The programs that run on it.

Windows, Chrome, Word, games.

PART 02 · PROCESSING

# THE CPU

### Control Unit

**Directs the work.**

Fetches each instruction and tells the parts what to do.

### ALU

**Math and logic.**

Adds, subtracts and compares numbers.

### Registers

**Tiny, very fast storage.**

Hold the values in use right now.

PART 02 · MEMORY

# RAM VS ROM

| | RAM<br>Random Access Memory | ROM<br>Read-Only Memory |
|---|---|---|
| POWER OFF | Loses its data | Keeps its data |
| ACCESS | Read and write | Read only |
| HOLDS | Open programs | Start-up code |

PART 02 · MEASURING DATA

# UNITS OF DATA

| UNIT | EQUALS |
|---|---|
| Bit | 0 or 1 |
| Nibble | 4 bits |
| Byte | 8 bits |
| Kilobyte (KB) | 1024 bytes |
| Megabyte (MB) | 1024 KB |
| Gigabyte (GB) | 1024 MB |
| Terabyte (TB) | 1024 GB |

**8**

BITS IN A BYTE

**1024**

EACH STEP UP

SMALLER UNIT: MULTIPLY · BIGGER UNIT: DIVIDE

# CONVERTING UNITS

**2048**

BYTES IN 2 KB

2 × 1024

**4096**

MB IN 4 GB

4 × 1024

**24**

BITS IN 3 BYTES

3 × 8

**2**

BYTES IN 16 BITS

16 ÷ 8

PART 03 · SECTION 02 RECAP

# NUMBER SYSTEMS

Four bases, one idea: every digit has a weight.

PART 03 · THE BASE IS HOW MANY DIGITS

# FOUR SYSTEMS

**2**
BINARY
0 and 1

**8**
OCTAL
0 to 7

**10**
DECIMAL
0 to 9

**16**
HEX
0 to 9, A to F

| A 10 | B 11 | C 12 | D 13 | E 14 | F 15 |
|---|---|---|---|---|---|

WEIGHT = BASE ^ POSITION, FROM 0 AT THE RIGHT

# DIGIT WEIGHTS

DECIMAL 352

| 3 | 5 | 2 |
|---|---|---|
| $\times 10^2 = 300$ | $\times 10^1 = 50$ | $\times 10^0 = 2$ |

$= 352$

BINARY 1011

| 1 | 0 | 1 | 1 |
|---|---|---|---|
| $\times 2^3 = 8$ | $\times 2^2 = 0$ | $\times 2^1 = 2$ | $\times 2^0 = 1$ |

$= 11$

PART 03 · LEARN IT BY HEART

# 0 TO 15

| DECIMAL | BINARY | OCTAL | HEX |
|---|---|---|---|
| 0 | 0000 | 0 | 0 |
| 1 | 0001 | 1 | 1 |
| 2 | 0010 | 2 | 2 |
| 3 | 0011 | 3 | 3 |
| 4 | 0100 | 4 | 4 |
| 5 | 0101 | 5 | 5 |
| 6 | 0110 | 6 | 6 |
| 7 | 0111 | 7 | 7 |

| DECIMAL | BINARY | OCTAL | HEX |
|---|---|---|---|
| 8 | 1000 | 10 | 8 |
| 9 | 1001 | 11 | 9 |
| 10 | 1010 | 12 | A |
| 11 | 1011 | 13 | B |
| 12 | 1100 | 14 | C |
| 13 | 1101 | 15 | D |
| 14 | 1110 | 16 | E |
| 15 | 1111 | 17 | F |

MULTIPLY BY THE WEIGHT, THEN ADD · (1011.101) BINARY

# ANY BASE TO DECIMAL

| 8 | 4 | 2 | 1 | | 0.5 | 0.25 | 0.125 |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | . | 1 | 0 | 1 |

**8 + 2 + 1 + 0.5 + 0.125** **= 11.625**

PART 03 · SAME METHOD

# OCTAL, HEX TO DECIMAL

## $(257)_8$

$2 \times 8^2 = 128$

$5 \times 8^1 = 40$

$7 \times 8^0 = 7$

**= 175**

## $(2AF)_{16}$

$2 \times 16^2 = 512$

$A: 10 \times 16^1 = 160$

$F: 15 \times 16^0 = 15$

**= 687**

DECIMAL TO ANY BASE · READ UP

# DIVIDE BY THE BASE

175 ÷ 8 · TO OCTAL

| STEP | QUOTIENT | DIGIT |
|---|---|---|
| 175 ÷ 8 | 21 | 7 |
| 21 ÷ 8 | 2 | 5 |
| 2 ÷ 8 | 0 | 2 |

ANSWER $(257)_8$

687 ÷ 16 · TO HEX

| STEP | QUOTIENT | DIGIT |
|---|---|---|
| 687 ÷ 16 | 42 | 15 = F |
| 42 ÷ 16 | 2 | 10 = A |
| 2 ÷ 16 | 0 | 2 |

ANSWER $(2AF)_{16}$

AFTER THE POINT · READ DOWN

# MULTIPLY BY THE BASE

0.3125 × 8 · TO OCTAL

| | |
|---|---|
| 0.3125 × 8 = 2.5 | 2 |
| 0.5 × 8 = 4.0 | 4 |

$(0.24)_8$

0.6875 × 16 · TO HEX

| | |
|---|---|
| 0.6875 × 16 = 11.0 | B |

$(0.B)_{16}$

GROUPS OF 3 BITS, FROM THE RIGHT

# BINARY TO OCTAL

$(10110101)_2$ → 010 (2) | 110 (6) | 101 (5) → 265

| 000 | 001 | 010 | 011 | 100 | 101 | 110 | 111 |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

GROUPS OF 4 BITS, FROM THE RIGHT

# BINARY TO HEX

$(10110101)_2$ → **1011** **0101**

**B** **5**

**B5**

Back

$(3F)_{16}$: 0011 1111 = $(111111)_2$

EXAMPLE: (1101.011) IN BINARY

# GROUPING WITH A POINT

OCTAL

001 1 | 101 5 . 011 3

15.3

HEX

1101 D . 0110 6

D.6

The rule

**Integer: pad zeros on the left. Fraction: pad zeros on the right.**

GO THROUGH BINARY · (765) IN OCTAL

# OCTAL TO HEX

01

### To binary

3 bits each: 111 110 101

02

### Regroup in 4s

0001 1111 0101

03

### To hex

1, F and 5

$(765)_8 = (1F5)_{16}$

Both = 501

PART 03 · SUMMARY

# WHICH METHOD WHEN?

| | | |
|---|---|---|
| 01 | **Any base to decimal** | Weights, then add |
| 02 | **Decimal to any base** | Divide integer, multiply fraction |
| 03 | **Binary and octal** | Groups of 3 bits |
| 04 | **Binary and hex** | Groups of 4 bits |
| 05 | **Octal and hex** | Go through binary |

PART 03 · WATCH OUT

# COMMON MISTAKES

## Grouping from the left

✗ `1011010 → 101 101 0`

✓ `1011010 → 001 011 010`

Start at the point, pad with zeros.

## 10 to 15 in hex

✗ `687 → 2 10 15`

✓ `687 → 2AF`

One symbol per digit: A = 10 … F = 15.

## Powers start at 1

✗ `101 → 8 + 0 + 2`

✓ `101 → 4 + 0 + 1`

The rightmost weight is 1, so 101 = 5.

Question 1 | 6 minutes

CONVERT TO DECIMAL, OCTAL AND HEX

# $(1101101)_2$

| DECIMAL | OCTAL | HEX |
| --- | --- | --- |
| ? | ? | ? |

ANSWER · QUESTION 1

# $(1101101)_2$

DECIMAL

**64 + 32 + 8 + 4 + 1**

**109**

OCTAL · GROUPS OF 3

001 101 101

**155**

HEX · GROUPS OF 4

0110 1101

**6D**

Question 2 | 6 minutes

CONVERT TO BINARY, THEN DECIMAL

# $(3B.8)_{16}$

BINARY ?

DECIMAL ?

ANSWER · QUESTION 2

# $(3B.8)_{16}$

BINARY · 4 BITS PER DIGIT

3 = 0011
B = 1011
8 = 1000

$(111011.1)_2$

DECIMAL · ADD THE WEIGHTS

**3 × 16 = 48**
**B = 11**
**8 ÷ 16 = 0.5**

$(59.5)_{10}$

SECTION 03 · REVIEW COMPLETE

# THANK YOU

**Questions? Ask now.**